

Task 1: Directory Navigation and Exploration

ls command:

This command is used to display all the files and directories of the current directory.

```
prajalm@prajal:~/Cloco$ ls
MyFile.txt  MyFolder
```

ls command with -a flag:

-a stands for "all". This flag is used to list all the files of the directory (i.e. normal and hidden) files

```
prajalm@prajal:~/Cloco$ ls -a
.  ..  .hidden.txt  MyFile.txt  MyFolder
```

ls command with -l flag:

-l stands for long format. This flag is used to list files in long list format with more details such as read/write permissions, created date, size, etc.

```
prajalm@prajal:~/Cloco$ ls -l
total 4
-rw-r--r-- 1 prajalm prajalm    0 Jun 17 10:51 MyFile.txt
drwxr-xr-x 2 prajalm prajalm 4096 Jun 17 10:51 MyFolder
```

ls command with -la flag:

-la means combination of -a and -l flag. This command will list all the files in long list format.

```
prajalm@prajal:~/Cloco$ ls -la
total 12
drwxr-xr-x  3 prajalm prajalm 4096 Jun 17 10:51 .
drwx----- 28 prajalm prajalm 4096 Jun 17 10:50 ..
-rw-r--r--  1 prajalm prajalm    0 Jun 17 10:51 .hidden.txt
-rw-r--r--  1 prajalm prajalm    0 Jun 17 10:51 MyFile.txt
drwxr-xr-x  2 prajalm prajalm 4096 Jun 17 10:51 MyFolder
```

Differences between -a, -l, -la flags:

The normal ls command only lists names of the files and directories. Flags are used to format the ls command.

-a flag is used to display all the files and directories names including the hidden ones.

-l flag is used to display all the files and directories names in long format.

-la is the combination of -a and -l. It lists all the files (including hidden files) in long format.

Task 2: Creating and managing directories

```
prajalm@prajal:~/Cloco$ mkdir fellowship_projects
prajalm@prajal:~/Cloco$ cd fellowship_projects/
prajalm@prajal:~/Cloco/fellowship_projects$ mkdir docs images
prajalm@prajal:~/Cloco/fellowship_projects$ ls
docs  images
prajalm@prajal:~/Cloco/fellowship_projects$ cd docs
prajalm@prajal:~/Cloco/fellowship_projects/docs$ touch notex.txt
prajalm@prajal:~/Cloco/fellowship_projects/docs$ ls
notex.txt
prajalm@prajal:~/Cloco/fellowship_projects/docs$ cd ..
prajalm@prajal:~/Cloco/fellowship_projects$ ls -l
total 8
drwxr-xr-x 2 prajalm prajalm 4096 Jun 17 11:12 docs
drwxr-xr-x 2 prajalm prajalm 4096 Jun 17 11:11 images
prajalm@prajal:~/Cloco/fellowship_projects$
```

Touch command:

Touch command is used to create a new file.

Syntax: touch <filename>.<extension>

Task 3: File Permission Basics

```
prajalm@prajal:~/Cloco/fellowship_projects/docs$ ls -l
total 0
-rw-r--r-- 1 prajalm prajalm 0 Jun 17 11:22 notes.txt
prajalm@prajal:~/Cloco/fellowship_projects/docs$ chmod a-w notes.txt
prajalm@prajal:~/Cloco/fellowship_projects/docs$ ls -l
total 0
-r--r--r-- 1 prajalm prajalm 0 Jun 17 11:22 notes.txt
prajalm@prajal:~/Cloco/fellowship_projects/docs$ chmod o+w notes.txt
prajalm@prajal:~/Cloco/fellowship_projects/docs$ ls -l
total 0
-r--r--rw- 1 prajalm prajalm 0 Jun 17 11:22 notes.txt
prajalm@prajal:~/Cloco/fellowship_projects/docs$
```

chmod command is used to change the permission of the file.

Syntax: `chmod <permission> <filename>`

Representation:

Group:	g
All:	a
Owner:	u
Other:	o

Read:	r
Write:	w
Execute:	x

To add permission: +,

To remove permission: -

a-w command means we are removing *write* permission from *all* users.

Here, 'a' means all, '-' means remove, 'w' means write permission.

u+w command means we are adding write permission for the user (owner).

Here: 'u' means user, '+' means add, 'w' means write permission.

Task 4: Numeric Permissions with chmod

```
prajalm@prajal:~/Cloco/fellowship_projects/docs$ ls -l
total 0
-r--r--rw- 1 prajalm prajalm 0 Jun 17 11:22 notes.txt
prajalm@prajal:~/Cloco/fellowship_projects/docs$ chmod 754 notes.txt
prajalm@prajal:~/Cloco/fellowship_projects/docs$ ls -l
total 0
-rwxr-xr-- 1 prajalm prajalm 0 Jun 17 11:22 notes.txt
prajalm@prajal:~/Cloco/fellowship_projects/docs$ chmod 660 notes.txt
prajalm@prajal:~/Cloco/fellowship_projects/docs$ ls -l
total 0
-rw-rw---- 1 prajalm prajalm 0 Jun 17 11:22 notes.txt
```

We can use the decimal form of the binary digit for read, write, and execute.

r	w	e		decimal
0	0	0	=	0
0	0	1	=	1
0	1	0	=	2
0	1	1	=	3
1	0	0	=	4
1	0	1	=	5
1	1	0	=	6
1	1	1	=	7

We can use the combination of these number to set permission for owner, group and others.

Eg: 754 means 7 for owner i.e read,write,execute,
5 for group i.e read and execute,
4 for others i.e read only.

Task 5: Ownership and Group Commands

```
prajalm@prajal:~/Cloco/fellowship_projects$ ls -l
total 8
drwxr-xr-x 2 prajalm prajalm 4096 Jun 17 11:22 docs
drwxr-xr-x 2 prajalm prajalm 4096 Jun 17 11:19 images
prajalm@prajal:~/Cloco/fellowship_projects$ sudo chown test docs
prajalm@prajal:~/Cloco/fellowship_projects$ ls -l
total 8
drwxr-xr-x 2 test      prajalm 4096 Jun 17 11:22 docs
drwxr-xr-x 2 prajalm prajalm 4096 Jun 17 11:19 images
prajalm@prajal:~/Cloco/fellowship_projects$ sudo chgrp test123 docs
prajalm@prajal:~/Cloco/fellowship_projects$ ls -l
total 8
drwxr-xr-x 2 test      test123 4096 Jun 17 11:22 docs
drwxr-xr-x 2 prajalm prajalm 4096 Jun 17 11:19 images
```

Before changing anything, the docs directory has prajalm as owner and group prajalm.
In above picture,

```
drwxr-xr-x 2 prajalm prajalm 4096 Jun 17 11:22 docs
              (owner)  (group)
```

chown: chown command is used to change the owner of the file or directory.

Syntax: chown <group_name> <file_name>

chgrp: chgrp command is used to change the group of the file or directory.

Syntax: chown <group_name> <file_name>

(if no group name is provided then it will change to user's current group)

Task 6: Practical Application

```
prajalm@prajal:~/Cloco$ mkdir fellowship_projects/backup
prajalm@prajal:~/Cloco$ ls fellowship_projects/backup/
prajalm@prajal:~/Cloco$ cp fellowship_projects/docs/notes.txt fellowship_projects/backup/
prajalm@prajal:~/Cloco$ ls fellowship_projects/backup/
notes.txt
prajalm@prajal:~/Cloco$ chmod 444 fellowship_projects/backup/notes.txt
prajalm@prajal:~/Cloco$ ls -l fellowship_projects/backup/notes.txt
-r--r--r-- 1 prajalm prajalm 0 Jun 17 13:21 fellowship_projects/backup/notes.txt
```

What we did:

- Made a backup folder
- Added a file for backup to the folder.
- Changed the permission of the folder to read only.

In real world scenario, we would want to have backup for critical files such as notes, logs, etc such that it cannot be modified or removed. In this case we would set permission of the file or directory to 444 (i.e read only for everyone)

Task 7: Advance file exploration

1. ls -R

The -R flag in the ls command represents recursive listing. It will recursively list the files and folders i.e. files and folders inside a folder will also be listed.

2. ls -t

It will list the files by sorting acc to file's modified time.

3. tree

This command will list all the files and folders in tree format.

```
prajalm@prajal:~/Cloco/fellowship_projects$ tree
.
├── backup
├── docs
│   └── notes.txt
└── images
    └── image.jpg

4 directories, 2 files
```

4. find

This command is used to find the files with filters such as name, size, type, etc.

Find all .txt files:

```
prajalm@prajal:~/Cloco/fellowship_projects$ find . -name "*.txt"
./docs/notes.txt
./backup/notes.txt
```

Find all file of size less than 5mb:

```
prajalm@prajal:~/Cloco/fellowship_projects$ find . -size -5M
.
./docs
./docs/notes.txt
./images
./images/image.jpg
./backup
```

5. du

DU stands for Disk Usage. This command is used to find the size of the directories.

```
prajalm@prajal:~/Cloco/fellowship_projects$ ls
backup docs images
prajalm@prajal:~/Cloco/fellowship_projects$ du docs
4      docs
prajalm@prajal:~/Cloco/fellowship_projects$ du docs -h
4.0K   docs
```

Task 8: Symbolic Links and Recursive Permissions

Symbolic links are the links that point to the original file from another directory. We use `ln` command to create a symbolic link.

-s flag: It will create a *symbolic link*

-h flag: It will create a *hard link*

```
prajalm@prajal:~/Cloco$ ln -s fellowship_projects/docs/notes.txt notes_link
prajalm@prajal:~/Cloco$ ls -l
total 8
drwxr-xr-x 5 prajalm prajalm 4096 Jun 17 13:20 fellowship_projects
-rw-r--r-- 1 prajalm prajalm    0 Jun 17 10:51 MyFile.txt
drwxr-xr-x 2 prajalm prajalm 4096 Jun 17 10:51 MyFolder
lrwxrwxrwx 1 prajalm prajalm   34 Jun 17 14:28 notes_link -> fellowship_projects/docs/notes.txt
```

As we can, the `notes_link` is pointing to `notes.txt` file inside `fellowship_projects/docs/` directory.

Creating archive folder:

```
prajalm@prajal:~/Cloco$ sudo chmod -R 660 fellowship_projects/archives
```

```
prajalm@prajal:~/Cloco$ sudo ls -lR -l
.:
total 8
drwxr-xr-x 6 prajalm prajalm 4096 Jun 17 14:30 fellowship_projects
-rw-r--r-- 1 prajalm prajalm    0 Jun 17 10:51 MyFile.txt
drwxr-xr-x 2 prajalm prajalm 4096 Jun 17 10:51 MyFolder
lrwxrwxrwx 1 prajalm prajalm   34 Jun 17 14:28 notes_link -> fellowship_projects/docs/notes.txt

./fellowship_projects/archives:
total 0
-rw-rw---- 1 root root 0 Jun 17 14:33 file1.txt
-rw-rw---- 1 root root 0 Jun 17 14:33 file2.txt
```

Difference between Softlink and Hardlink:

- A symbolic link is a shortcut that points to another file in different path.
- If the original file is deleted or moved, the link breaks
- They are used to have easier access to deeper/nested files in the filesystem.
- A hard link is another name for the same file on disk.
- It shares the same inode and remains valid even if the original is deleted.
- It is mainly used for managing files, creating backups.

Task 9: Advanced Archiving and Linux Package Management

Archiving:

Archiving is the process of combining multiple files and directories in one file.

Compressing:

Compressing is the process of reducing the file size

Archiving files/folder

Use command `tar -cvf fellowship_archive.tar fellowship_projects` to archive the folder 'fellowship_projects'.

Here, -c flag means 'create' a new archive,
-v means 'verbose' which simply tells to show progress
-f flag is used to give name to our archive file.

```
prajalm@prajal:~/ClOCO$ sudo tar -cvf fellowship_archive.tar fellowship_projects
fellowship_projects/
fellowship_projects/docs/
fellowship_projects/docs/notes.txt
fellowship_projects/archives/
fellowship_projects/archives/file2.txt
fellowship_projects/archives/file1.txt
fellowship_projects/images/
fellowship_projects/images/image.jpg
fellowship_projects/backup/
fellowship_projects/backup/notes.txt
```

Compressing the archive:

We can compress the existing tar file using the zip command:

```
prajalm@prajal:~/ClOCO$ sudo zip -e fellowship_projects_archive.zip fellowship_projects_archive.tar
Enter password:
Verify password:
  adding: fellowship_projects_archive.tar (deflated 97%)
```

The -e flag will enable password to unzip the zip file.

We can also add additional flags to compress the files while archiving.

```
prajalm@prajal:~/ClOCO$ sudo tar -cvzf fellowship_archive_compressed.tar fellowship_projects
fellowship_projects/
fellowship_projects/docs/
fellowship_projects/docs/notes.txt
fellowship_projects/archives/
fellowship_projects/archives/file2.txt
fellowship_projects/archives/file1.txt
```

Following are the detail of the compression flag:

Format	Flag	File Extension	Speed	Compression
gzip	-z	.tar.gz	Fastest	Moderate
bzip2	-j	.tar.bz2	Slower	Better
xz	-J	.tar.xz	Slowest	Best

Here is the detailed result of compression using different flags. We can see size difference in all of them:

```
prajalm@prajal:~/Cloco$ ls -lh
total 24K
-rw-r--r-- 1 prajalm prajalm 190 Jun 17 15:47 fellow_archive.zip
-rw-r--r-- 1 root root 359 Jun 17 15:58 fellowship_archive_compressed.tar.bz2
-rw-r--r-- 1 root root 358 Jun 17 15:58 fellowship_archive_compressed.tar.gz
-rw-r--r-- 1 root root 372 Jun 17 15:56 fellowship_archive_compressed.tar.xz
drwxr-xr-x 6 prajalm prajalm 4.0K Jun 17 14:30 fellowship_projects
-rw-r--r-- 1 prajalm prajalm 0 Jun 17 10:51 MyFile.txt
drwxr-xr-x 2 prajalm prajalm 4.0K Jun 17 10:51 MyFolder
lrwxrwxrwx 1 prajalm prajalm 34 Jun 17 14:28 notes_link -> fellowship_projects/docs/notes.txt
```

Unarchiving the archived file:

Unarchiving means extracting the files from the archived file. It is the opposite of archiving.

```
prajalm@prajal:~/Cloco$ sudo tar -xvf fellowship_archive_compressed.tar
fellowship_projects/
fellowship_projects/docs/
fellowship_projects/docs/notes.txt
fellowship_projects/archives/
fellowship_projects/archives/file2.txt
fellowship_projects/archives/file1.txt
fellowship_projects/images/
fellowship_projects/images/image.jpg
fellowship_projects/backup/
fellowship_projects/backup/notes.txt
```

Unzipping the zipped file:

We can use the 'unzip' command to unzip the compressed zip file.

```
prajalm@prajal:~/Cloco$ unzip fellowship_projects_archive.zip
Archive:  fellowship_projects_archive.zip
[fellowship_projects_archive.zip] fellowship_projects_archive.tar password:
replace fellowship_projects_archive.tar? [y]es, [n]o, [A]ll, [N]one, [r]ename: n
```

PART B: ADVANCED PACKAGE MANAGEMENT

Installing nmap with apt

```
prajalm@prajal:~/Cloco$ nmap --version
Nmap version 7.93 ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.3.6 openssl-3.0.16 libssh2-1.10.0 libz-1.2.13 libpcap-1.10.3
Compiled without:
Available nsock engines: epoll poll select
prajalm@prajal:~/Cloco$ nmap localhost
Starting Nmap 7.93 ( https://nmap.org ) at 2025-06-17 16:39 +0545
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000072s latency).
Other addresses for localhost (not scanned): ::1
Not shown: 995 closed tcp ports (conn-refused)
PORT      STATE SERVICE
80/tcp    open  http
631/tcp   open  ipp
3306/tcp  open  mysql
5432/tcp  open  postgresql
8000/tcp  open  http-alt

Nmap done: 1 IP address (1 host up) scanned in 0.03 seconds
```

Installing using snap:

```
prajalm@prajal:~/Cloco$ sudo snap install nmap
nmap 7.95 from Maximiliano Bertacchini (maxibertao) installed
```

Difference between apt, dpkg, snap

apt	dpkg	snap
High-level package manager for Debian-based systems.	Low-level tool to install .deb packages manually.	Cross-distro package manager.
used for installing packages from official repositories	best for installing local .deb files.	It is useful for universal app deployment.
Resolves dependencies automatically	Does not resolve dependencies.	It includes all dependencies